EE656 Project Report on Decoupling Identity Confounders for Enhanced Facial Expression Recognition

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June 2026

Abstract

Facial expression recognition is a challenging computer vision task due to large intra-class variations which often interfere with expression-related features. In this project, we implement and evaluate DICE-FER, an enhanced facial expression recognition framework devised by **Aquib et al.(2025)** .The model employs separate expression and identity encoders, global and local mutual information estimation networks and adversarial learning to effectively disentangle two representations. The implementation is trained and evaluated on benchmark FER datasets following pre-processing and augmentation as described in the original paper. Performance is evaluated using classification accuracy, precision, recall, F1 score, confusion matrix.

1. Introduction

Facial expression recognition is an important task in computer vision that aims to identify emotions from facial images. It has many real-world applications in the field of healthcare, surveillance, driver monitoring systems. As facial expressions exhibit large intra-class variations, FER remains a challenging problem. Factors such as facial identity, head pose, illumination makes things more complex.

One major challenge in FER is the presence of identity-specific information in facial representations. Use of conventional methods resulted in learning features that contain both expression and identity information. Thus the learned representations were not effectively made on unseen people.

DICE-FER works on this problem by learning expressions which are not related with identity through mutual information optimizations. Instead of reconstructing images, the framework maximizes mutual information between images sharing same facial expression and also minimizes mutual information between expression and identity features using adversarial learning. This enables model to effectively learn facial expressions without mixing them with identity of a person.

In this project, we implement this DICE-FER framework. The implementation follows the methodology used in the original paper.

2. Theoretical Base

The DICE-FER framework is based on Information Theory and Mutual information estimation. The objective is to learn facial representations in which expression-related information is preserved while identity-related information is removed. In this section, we wrote some theoretical concepts used in the proposed methods. The theoretical background follows from the **section 3, 4, 5.4** of the original DICE-FER paper.

2.1. Mutual Information

Mutual Information (MI) measures statistical dependency between two random variables. It shows how much information one variable provides about another. For two random variables X and Y, the mutual information is defined as

$$I(X;Y) = \int_{\mathcal{X}} \int_{\mathcal{Y}} p(x,y) \log \left(\frac{p(x,y)}{p(x)p(y)} \right) \, dx \, dy \tag{1}$$

where

- $p(x, y)$ is the joint probability distribution,
- $p(x)$ and $p(y)$ are the marginal probability distributions.

Higher the value of MI, stronger the dependence between two variables.

MI can also be measured as KL divergence between joint distribution and product of marginal distributions,

$$I(X;Y) = D_{KL}(P_{XY} \,\|\, P_X P_Y) \tag{2}$$

2.2. Mutual Information Neural Estimation (MINE)

For high-dimensional image representations, direct computation of MI is not feasible. Hence DICE-FER employs MINE which estimates MI using neural statistics network. Using Donsker-Vardhan representation, MINE is

$$\hat{I}_{DV}(X,Y) = \mathbb{E}_{p(x,y)}[T_\theta(X,Y)] - \log\Big(\mathbb{E}_{p(x)p(y)}\Big[e^{T_\theta(X,Y)}\Big]\Big) \tag{3}$$

where T_θ denotes a neural statistics network parametrized by θ. It estimates MI directly from image features.

2.3. Adversarial Mutual Information Minimization

To prevent identity information from leaking into expression representation, DICE-FER introduces adversarial discriminator. The discriminator distinguish between

- joint samples (E, I),
- independently sampled expression and identity representations.

The adversarial loss is

$$L_{adv} = \mathbb{E}_{p(E)p(I)}[\log D(E, I)] + \mathbb{E}_{p(E,I)}[\log(1 - D(E, I))] \tag{4}$$

The identity learning objective becomes

$$L_{id} = L_{MI}^{id} - \zeta_{adv} L_{adv}, \tag{5}$$

where ζ_{adv} controls strength of disentanglement.

3. Proposed Methodology

We aim to disentangle facial expression information from identity-related features using MI maximization and adversarial learning.

The complete training procedure consists of three stages:

1. Expression representation learning
2. Identity representation learning
3. Facial expression classification

Throughout training, two images having same expression but belonging to different indentities were selected as input. Denote this images by M and N. Framework learns two latent representations:

- Expression embedding:
$$E_M,\ E_N \in \mathbb{R}^{64}$$
- Identity embedding:
$$I_M,\ I_N \in \mathbb{R}^{64}$$

Final objective is to maximize expression-related information and minimize dependence between and expression and identity representations.

3.1. Stage 1: Expression representation learning

During first stage, paired images are passed through a shared expression encoder based on ResNet-18. The encoder extracts both global and local feature representations, which are used to estimate MI between the paired images. Since both images depict the same facial expression, their expression representations should contain highly similar information despite differences in identity. To achieve this, Donsker–Varadhan Mutual Information Neural Estimation (MINE) is employed using both global and local statistics networks. The expression objective maximizes

$$I(E_M, E_N),$$

which encourages the encoder to retain only expression-related information. To further reduce variation between paired embeddings, an additional L1 consistency loss is introduced

$$\mathcal{L}_{L1} = \|E_M - E_N\|_1.$$

The complete expression learning objective is therefore

$$\mathcal{L}_{exp} = -\mathcal{L}_{MI}^{exp} + \delta\mathcal{L}_{L1},$$

3.2. Stage 2: Identity representation learning

The identity representation learning branch is responsible for learning features that are unique to an individual's facial identity while minimizing the influence of facial expressions. Since every person has distinct facial characteristics, these identity-specific features can interfere with expression recognition if they are not handled separately. To address this, the identity encoder learns a representation that preserves identity information through mutual information maximization. This helps the network distinguish identity-related features from expression-related features, allowing the expression recognition module to focus on emotion rather than the person's appearance.

3.3. Stage 3: Expression Classification

After learning an identity-invariant expression representation, the extracted features are passed to the expression classifier to predict the corresponding facial expression. The classifier consists of a fully connected layer followed by a softmax activation function, which produces the probability distribution over the expression classes. During training, the cross-entropy loss is used to optimize the classifier by encouraging correct predictions while penalizing incorrect ones. By relying on expression features that are less affected by identity information, the classifier is able to achieve more reliable and robust facial expression recognition.

4. Experimental Setup

4.1. Dataset

The original DICE-FER framework was evaluated on four benchmark FER datsets: CK+, Oulu-Casia, RAF-DB, and AffectNet. Owing to dataset availability and project time constraints, our replication experiments were conducted using RAF-DB and CK+ Datsets. RAF-DB was publicly available and was used after following the preprocessing pipeline given in original paper.

For CK+ dataset, the official dataset distribution through author was discontinued in 2025. There was no publicly available original version. So we utilized a publicly available preprocessed CK+ dataset and reconstructed it to be compatible on DICE-FER. The contruction of CK+ is explained in section 4.2 of this report.

The original paper also reports experiments on the Oulu-CASIA and AffectNet datasets. However, these datasets were not publicly available and required submission of an access request to the dataset authors. The approval process typically required approximately one week. As this waiting period exceeded the timeline allocated for the course project, these datasets were not used in our replication experiments. Consequently, all the reported experimental results in this work are based solely on the RAF-DB and reconstructed CK+ datasets.

4.2. Data Pre-processing

All facial images were converted into a standardized format before training. Images from both RAF-DB and CK+ were resized to 112x112 pixels and converted to grayscale. For unaligned faces, facial alignement was performed using Multi-task cascaded convolutional network (MTCNN), similar to the paper.

For the CK+ dataset, in the reconstruction process we selected only peak expression frames for the seven expression classes used in original work. We resized the image from 48x48 pixels to 112x112 pixels using bicubic interpolation to match input size expected by ResNet-18 backbone. Image whitening was done and subject identities were reconstructed by assigning images to approximately 123 subjects on basis of ordering provided in metadata. A structured dataset containing image paths, expression labels, identity labels was created for training purpose.

For the RAF-DB dataset, the provided train-test partition was kept as it is. Each image was resized to 112x112, converted to grayscale, whitened, organized according to its expression label.

A validation stage was also performed to ensure all image paths were valid, expression labels were correctly assigned and identities were properly reconstructed.

4.3. Data Augmentation

Data augmentation was applied to increase the diversity of the training data. For each training image, we generated 14 augmented samples by combining seven rotation angles $\{-15°, -10°, -5°, 0°, 5°, 10°, 15°\}$, with horizontal flipping. The augmentation process was applied only to the training set, while the validation and testing sets remained unmodified.

4.4. Hardware and Software

All experiments were implemented using Python and PyTorch framework. Data preprocessing utilized OpenCV, MTCNN, Pillow, Numpy, Scikit-learn. Model training was done on Nvidia L4 via cloud GPU.

5. Results

5.1. RAF-DB

Figure 1 shows our replication result on the RAF-DB dataset. The model achieved an overall classification accuracy of **82.72%**, with a macro precision of **0.752**, recall of **0.741**, and F1-score of **0.744**.

Method	Setting	Expressions	Accuracy (%)	Precision	Recall	F1 Score
DICE-FER (ours)	image-based	7	82.72	0.752	0.741	0.744

Figure 1: table

Confusion Matrix analysis: Figure 2 shows the normalized confusion matrix obtained on the RAF-DB test set. As expected, the strongest responses occur along the diagonal, indicating that our model correctly recognizes most facial expressions.

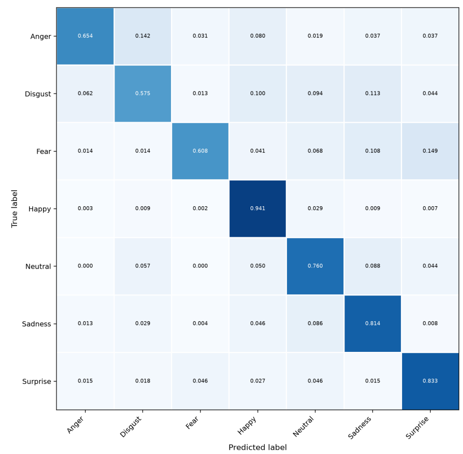

Figure 2: Confusion Matrix

Among all expression categories, *Happy* achieved the highest recognition accuracy, with approximately **94.1%** of happy expressions correctly classified. This suggests that happiness exhibits highly distinctive facial characteristics that are effectively captured by the learned expression representation.

The model also performed well on *Surprise* (**83.3%**), *Sadness* (**81.4%**), and *Neutral* (**76.0%**), indicating that these expressions possess relatively discriminative visual patterns.

On the other hand, *Disgust* (**57.5%**), *Fear* (**60.8%**), and *Anger* (**65.4%**) were the most challenging classes. These emotions showed significant confusion with visually similar expressions. Particularly, anger was frequently misclassified as disgust and happy, while disgust was confused with sadness and happy. Fear showed a notable confusion with surprise and sadness, reflecting the substantial overlap in facial muscle activations among these emotions.

These misclassification patterns are consistent with previous FER literature, where negative facial expressions generally exhibit greater intra-class variability and smaller inter-class margins than positive expressions.

Compared with the original DICE-FER implementation, which reported an accuracy of **90.30%** on RAF-DB, our implementation achieved **82.72%**. Despite the reduction in overall accuracy, the confusion matrix demonstrates that the replicated model successfully learned meaningful expression representations, particularly for highly distinguishable expressions such as happiness and surprise. The remaining performance gap is likely due to implementation differences, preprocessing variations, computational constraints, and the inherent difficulty of reproducing adversarial mutual-information optimization

5.2. CK+

Figure 3 shows the average performance across 10-fold cross validation of our replicated model on CK+ Dataset.

	Mean	Std
Accuracy	92.3957	5.7493
Precision	0.7763	0.1109
Recall	0.7734	0.1302
F1	0.769	0.1194

Figure 3: Enter Caption

Figure 4 shows the average performance of model using radar chart. Among the four evaluation metrics, accuracy achieved the highest value, while precision, recall and F1 score remained closely aligned. This indicates a balanced classification performance without bias towards any particular metric.

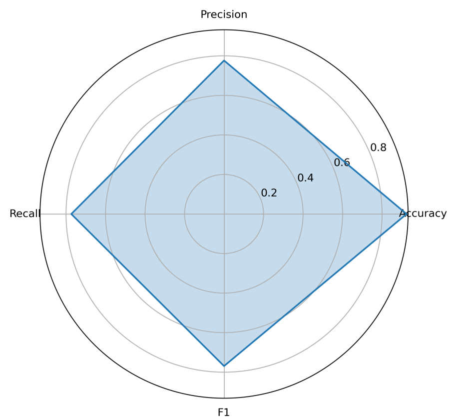

Figure 4: Enter Caption

Figure 5 shows the distribution of accuracy across the 10 folds.

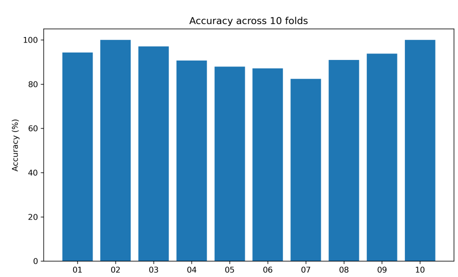

Figure 5: Enter Caption

6. Limitations

Although the proposed DICE-FER framework was successfully implemented, we faced several practical limitations that prevented an exact replication of the results reported in the original paper. One of the main challenges was dataset availability. The official CK+ dataset was no longer publicly available, so we reconstructed a compatible version using a publicly available preprocessed dataset. Additionally, the Oulu-CASIA and AffectNet datasets required prior approval from the dataset authors, making them unavailable within the project timeline.

The project was completed within a limited time which restricted extensive hyperparameter tuning, repeated experiments, and detailed analysis. Computational limitations also reduced the number of training runs that could be performed for such a complex architecture involving multiple encoders, mutual information estimation networks, and adversarial optimization.

Finally, reproducing mutual-information-based adversarial learning is inherently challenging, as the final performance is sensitive to implementation details, initialization, optimization strategy, and random sampling. Most of the implementation, dataset preparation, debugging, and experimentation was also carried out by a single contributor, which further limited the scope for extensive validation. Despite these challenges, we replicated implementation successfully and reproduced the core methodology of DICE-FER and achieved competitive performance on both the RAF-DB and CK+ datasets.

7. Proposed Extensions

Building on the DICE-FER framework, we identify two extensions that can further improve recognition accuracy and training stability. These extensions are proposed as theoretical extensions based on deep learning techniques and we didn't experimentally tested it evaluated it due to project time constraints.

- **Label Smoothing:** The original DICE-FER trains expression classifier using standard cross-entropy loss. In FER, several emotions share similar facial movements making it hard to distinguish them. For example, fear and surprise shows similar eyebrows and mouth movements, or sadness and neutral often have same facial appearance. We can employ label smoothing to tackle this. Instead of assigning a probability 1.0 to the ground-truth class, a small portion of probability mass is distributed among remaining classes. This can act as regularizer and improve model calibration and reduce overfitting.

- **Cosine Annealing:** As the original paper uses a fixed learning rate throughout training, our observation showed that MI objective converges rapidly during initail epochs, and then maintains a constant learning rate.

To obtain more stable convergence, a cosine annealing learning rate schedule can be adopted. This will make the learning rate decrease smoothly following a cosine decay. This might enable expression encoder to perform large updates during early training and fine adjustments near convergence, which will give improved optimization stability and better feature representations.

8. Conclusion

Our work presented the implementation and evaluation of DICE-FER. We attempted to replicate the model using a three-stage learning strategy comprising of expression representation learning, identity disentanglement, and supervised expression classification. Experiments were conducted on RAF-DB and CK+.

The design of DICE-FER provides a flexible foundation for future developments. We also proposed two theoretical enhancements to the model to improve performance and stability. Overall, our work highlights effectiveness of representation disentanglement for FER and provides a strong basis for further investigation into identity-invariant feature learning.

9. References

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